8.30

Since $\hat{p} = x/n = 42/500 = 0.084$,
$$z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n} = 1.96\sqrt{(0.084)(0.916)/500} = 0.024.$$
A 95% confidence interval for p is
$$\left(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}, \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\right) = (0.084 - 0.024, 0.084 + 0.024) = (0.060, 0.108).$$

8.33

(a)
$$n = \frac{1}{4}\left(\frac{z_{\alpha/2}}{E}\right)^2 = \frac{1}{4}\left(\frac{z_{0.025}}{E}\right)^2 = \frac{1}{4}\left(\frac{1.96}{0.1}\right)^2 = 96.04$$

Thus, the sample size should be at least 97.

(b)
$$n = p(1-p)\left(\frac{z_{\alpha/2}}{E}\right)^2 = p(1-p)\left(\frac{z_{0.025}}{E}\right)^2 = (0.2)(0.8)\left(\frac{1.96}{0.1}\right)^2 = 61.47$$

The sample size should be at least 62.

8.37

(a)
$$n = p(1-p)\left(\frac{z_{\alpha/2}}{E}\right)^2 = p(1-p)\left(\frac{z_{0.025}}{E}\right)^2 = (0.2)(0.8)\left(\frac{1.96}{0.01}\right)^2 = 6146.56$$

The required sample size is 6147.

(b)
Since $\hat{p} = x/n = 37/400 = 0.0925$,
$$z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n} = 1.96\sqrt{(0.0925)(0.9075)/400} = 0.0284.$$
A 95% confidence interval for p is
$$\left(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}, \hat{p} + z_{\alpha/2}\sqrt{\hat{p}(1-\hat{p})/n}\right) = (0.0925 - 0.0284, 0.0925 + 0.0284) = (0.0641, 0.1209).$$

8.39

(a) $H_0: p = 0.15$, $H_1: p < 0.15$
$\alpha = 0.05$
The test statistic is

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

The rejection region is $z \leq -z_\alpha = -z_{0.05} = -1.645$.
Since $\hat{p} = x/n = 19/140 = 0.1357$,

$$z = \frac{0.1357 - 0.15}{\sqrt{(0.15)(0.85)/140}} = -0.474 > -1.645$$

Do not reject H_0.

(b) p-value $= P(Z < -0.474) = 0.3178$

8.42

(a) $H_0: p = 0.7$, $H_1: p \neq 0.7$
$\alpha = 0.01$
The test statistic is

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

The rejection region is $|z| \geq z_{\alpha/2} = z_{0.005} = 2.58$.
Since $\hat{p} = x/n = 802/1180 = 0.6797$,

$$z = \frac{0.6797 - 0.7}{\sqrt{(0.7)(0.3)/1180}} = -1.5217$$

Since $|z| = 1.5217 < 2.58$, we do not reject H_0.

(b) p-value $= P(|Z| > 1.52) = 2\Phi(-1.52) = 2(0.064) = 0.128$

8.48

$$n = 14, s = 3.27$$

$$\left(\frac{(n-1)s^2}{\chi^2_{\frac{\alpha}{2},n-1}}, \frac{(n-1)s^2}{\chi^2_{1-\frac{\alpha}{2},n-1}}\right) = \left(\frac{(n-1)s^2}{\chi^2_{0.05,13}}, \frac{(n-1)s^2}{\chi^2_{0.955,13}}\right) = \left(\frac{13(3.27^2)}{22.362}, \frac{13(3.27^2)}{5.892}\right)$$
$$= (6.22, 23.59)$$

8.51

$n = 22,\ \bar{x} = 536.8,\ s = 105.2$

(a) $H_0: \mu = 511, H_1: \mu > 511$
$\alpha = 0.05$
Since $n = 22$ is small, the test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

The rejection region is $t \geq t_{\alpha,n-1} = t_{0.05,21} = 1.721$.

$$t = \frac{536.8 - 511}{105.2/\sqrt{22}} = 1.15 < 1.721$$

Do not reject H_0.

(b)

$$\left(s\sqrt{\frac{(n-1)}{\chi^2_{\frac{\alpha}{2},n-1}}}, s\sqrt{\frac{(n-1)}{\chi^2_{1-\frac{\alpha}{2},n-1}}}\right) = \left(s\sqrt{\frac{(n-1)}{\chi^2_{0.005,21}}}, s\sqrt{\frac{(n-1)}{\chi^2_{0.995,21}}}\right)$$
$$= \left(105.2\sqrt{\frac{21}{41.40}}, 105.2\sqrt{\frac{21}{8.03}}\right) = (74.9, 170.1)$$

8.53

$n = 13, \bar{x} = 79.69, s^2 = 27.56, s = 5.25$

(a) $H_0: \sigma = 10$, $H_1: \sigma \neq 10$

$\alpha = 0.05$

The test statistic is

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2}.$$

The rejection region is $\chi^2 \geq \chi^2_{\frac{\alpha}{2},n-1} = \chi^2_{0.025,12} = 23.337$ or $\chi^2 \leq \chi^2_{1-\frac{\alpha}{2},n-1} = \chi^2_{0.975,12} = 4.404$.

$$\chi^2 = \frac{12(27.56)}{10^2} = 3.307 < 4.404$$

Reject H_0.

(b) $H_0: \mu = 85$, $H_1: \mu \neq 85$

$\alpha = 0.05$

The test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

The rejection region is $|t| \geq t_{\alpha/2,n-1} = t_{0.025,12} = 2.179$.

$$t = \frac{79.69 - 85}{5.25/\sqrt{13}} = 3.647 > 2.719$$

Since $|t| = 3.647 > 2.719$, we reject H_0.

(c) p-value $= 2P(T > 3.647) = 2(0.0017) = 0.0034$

9.2

Both $m = 65, n = 60$ are large.

$$\bar{x} - \bar{y} \pm z_{\frac{\alpha}{2}}\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}} = 107 - 113 \pm z_{0.025}\sqrt{\frac{10^2}{65} + \frac{13^2}{60}} = -6 \pm 1.96(2.087)$$
$$= -6 \pm 4.1$$

A 95% confidence interval is $(-10.1, -1.9)$.

9.4

$m = n = 35$ are large.

$H_0: \mu_1 = \mu_2, H_1: \mu_1 > \mu_2$

$\alpha = 0.1$

The test statistic is

$$Z = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{m} + \frac{S_2^2}{n}}}.$$

The rejection region is $z \geq z_\alpha = z_{0.01} = 1.28$.

$$z = \frac{29700 - 25500}{\sqrt{\frac{(9700)^2}{35} + \frac{(7800)^2}{35}}} = 1.996 > 1.28$$

Reject H_0.

There is sufficient evidence that mean drying time of the paints made by Company A is shorter.

p-value $= P(Z > 1.996) = 1 - \Phi(1.996) = 1 - 0.977 = 0.023$

9.6

$m = n = 15$ are small. $\bar{x} = 34.6,\ s_1 = 2.5;\ \bar{y} = 37.2, s_2 = 2.2$

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{14(2.5)^2 + 14(2.2)^2}{15+15-2} = 5.545$$

Since $t_{\frac{\alpha}{2},m+n-2} = t_{0.025,28} = 2.048$,

$$\bar{x} - \bar{y} \pm t_{\frac{\alpha}{2},m+n-2} s_p \sqrt{\frac{1}{m} + \frac{1}{n}} = 34.6 - 37.2 \pm 2.048\sqrt{5.545}\sqrt{\frac{1}{15} + \frac{1}{15}} = -2.6 \pm 1.76.$$

Thus, a 95% confidence interval for $\mu_1 - \mu_2$ is $(-4.36, -0.84)$.

9.9

$m = n = 5$ are small. $\bar{x} = 32,\ s_1 = 8.63;\ \bar{y} = 29.2, s_2 = 7.12$

$H_0: \mu_1 = \mu_2, H_1: \mu_1 > \mu_2$

$\alpha = 0.1$

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{4(8.63)^2 + 4(7.12)^2}{5+5-2} = 62.59$$

The test statistic is

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{\frac{1}{m} + \frac{1}{n}}}.$$

The rejection region is $t \geq t_{\alpha,m+n-2} = t_{0.1,8} = 1.397.$

$$t = \frac{32 - 29.2}{\sqrt{62.59}\sqrt{\frac{1}{5} + \frac{1}{5}}} = 0.560 < 1.397$$

Do not reject H_0.

p-value$= P(T > 0.56) = 0.2954$

9.17

Both $m = 22,\ n = 20$ are small.

(a)

$$\nu = \frac{\left(\frac{s_1^2}{m}+\frac{s_2^2}{n}\right)^2}{\frac{\left(s_1^2/n_1\right)^2}{m-1}+\frac{\left(s_2^2/n_2\right)^2}{n-1}} = \frac{\left(\frac{(12.2)^2}{22}+\frac{(3.7)^2}{20}\right)^2}{\frac{\left((12.2)^2/22\right)^2}{21}+\frac{\left((3.7)^2/20\right)^2}{19}} = 25.18 \ \rightarrow \ \nu = 25$$

$$\bar{x}_1 - \bar{x}_2 \pm t_{0.05,25}\sqrt{\frac{s_1^2}{m}+\frac{s_2^2}{n}} = 55.7 - 52.8 \pm 1.708\sqrt{\frac{(12.2)^2}{22}+\frac{(3.7)^2}{20}}$$

$$= 2.9 \pm 4.66$$

Thus, a 90% CI is $(-1.76, 7.56)$.

(b) $H_0\colon \mu_1 - \mu_2 = 0,\ H_1\colon \mu_1 - \mu_2 > 0$

$\alpha = 0.1$

The test statistic is

$$T = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{S_1^2}{m}+\frac{S_2^2}{n}}}.$$

The rejection region is $t > t_{\alpha,\nu} = t_{0.1,25} = 1.316$.

$$t = \frac{55.7 - 52.8}{\sqrt{\frac{(12.2)^2}{22}+\frac{(3.7)^2}{20}}} = 1.062 < 1.316$$

Do not reject H_0. There is no significant evidence that the computer technology industry pays more than the automotive industry to recent college graduates.

9.18

	Person						
	1	2	3	4	5	6	7
Old model	17	22	20	14	15	21	24
New model	15	19	18	13	15	19	23
Difference	2	3	2	1	0	2	1

(a)

$\bar{d} = 1.57, s_d = 0.976$

$$\bar{d} \pm t_{\frac{\alpha}{2},n-1} \frac{s_d}{\sqrt{n}} = \bar{d} \pm t_{0.025,6} \frac{s_d}{\sqrt{6}} = 1.57 \pm 2.447 \frac{0.976}{\sqrt{7}} = 1.57 \pm 0.90$$

A 95% confidence interval is $(0.67, 2.47)$.

(b) $H_0: \delta = 0, H_1: \delta > 0$

$\alpha = 0.05$

The test statistic is

$$T = \frac{\bar{D}}{S_d/\sqrt{n}}.$$

The rejection region is $t \geq t_{\alpha,n-1} = t_{0.05,6} = 1.943$.

$$t = \frac{1.57}{0.976/\sqrt{7}} = 4.256 > 1.943$$

Reject H_0.

p-value$= P(T > 4.256) = 0.0027$

(c) $m = n = 7, \bar{x} = 19, s_1 = 3.74; \bar{y} = 17.4, s_2 = 3.36$

$H_0: \mu_1 - \mu_2 = 0, H_1: \mu_1 - \mu_2 > 0$

$$s_p^2 = \frac{(m-1)s_1^2 + (n-1)s_2^2}{m+n-2} = \frac{6(3.74)^2 + 6(3.36)^2}{7+7-2} = 12.64$$

The test statistic is

$$T = \frac{\bar{X} - \bar{Y}}{S_p\sqrt{\frac{1}{m} + \frac{1}{n}}}.$$

The rejection region is $t \geq t_{\alpha,m+n-2} = t_{0.05,12} = 1.782$.

$$t = \frac{19 - 17.4}{\sqrt{12.64}\sqrt{\frac{1}{7} + \frac{1}{7}}} = 0.8419 < 1.782$$

Do not reject H_0 .
p-value= $P(T > 0.8419) = 0.2082$
The decision is reversed because the within subject variation is ignored in part (c).

9.23

$n = 12,\ \overline{d} = -16.5,\ \ s_d = 10.7$

$H_0: \delta = 0, H_1: \delta < 0$

$\alpha = 0.05$

The test statistic is

$$T = \frac{\overline{D}}{S_d/\sqrt{n}}.$$

The rejection region is $t\ \leq -t_{\alpha,n-1} = -t_{0.05,11} = -1.796.$

$$t = \frac{-16.5}{10.7/\sqrt{12}} = -5.342 < -1.796$$

Reject H_0. There is sufficient evidence to conclude that the hypertension medicine dropped the blood pressure.

p-value= $P(T < -5.342) = 0.0001$